How journals can improve code and data sharing in Neuroscience

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Slides: http://bit.ly/eglen-neuromatch (CC-BY license)

Declarations

1. Member of *Bullied into Bad Science* campaign
2. Affiliate editor of *bioRxiv*
3. Senior editor of *Scientific Data*

Acknowledgements

1. Corina Logan and Laurent Gatto.
2. Joined by Ross Mounce, Adrian Curry and myself. Joined by Lauren Maggio.

http://bulliedintobadscience.org has most of the key background

Do ask questions!

Thanks to Ben Farrar and Yvonne Nobis for material.

Problems

- we are sharing the wrong stuff. Don't share the shiny paper, share the raw materials that go into the paper.

How do journals host code?

Eiko Fried
@EikoFried

Journal didn't know what to do with our #rstats supplementary .. so they took screenshots and pasted into word o.O

link.springer.com/article/10.100...

Electronic Supplementary Material

```
#################################################################
#                                                               #
#                   Supplementary Materials                     #
#                                                               #
#         Mental disorders as networks of problems:             #
#                 a review of recent insights                   #
#                                                               #
#              Fried, van Borkulo, et al., 2016                 #
#                                                               #
#       Social Psychiatry and Psychiatric Epidemiology          #
#                                                               #
#################################################################

library(qgraph)

##### Figure 1

### Figure 1A
AdjacencyMatrix.1A <- matrix(0,8,8)
AdjacencyMatrix.1A[1,2] <- AdjacencyMatrix.1A[2,1] <-
AdjacencyMatrix.1A[2,3] <- AdjacencyMatrix.1A[3,2] <-
AdjacencyMatrix.1A[2,4] <- AdjacencyMatrix.1A[4,2] <-
AdjacencyMatrix.1A[2,5] <- AdjacencyMatrix.1A[5,2] <-
AdjacencyMatrix.1A[3,4] <- AdjacencyMatrix.1A[4,3] <-
AdjacencyMatrix.1A[3,5] <- AdjacencyMatrix.1A[5,3] <-
AdjacencyMatrix.1A[3,6] <- AdjacencyMatrix.1A[6,3] <-
AdjacencyMatrix.1A[3,8] <- AdjacencyMatrix.1A[8,3] <-
AdjacencyMatrix.1A[4,7] <- AdjacencyMatrix.1A[7,4] <-
```

Opportunities

Sharing code now seen as important

- Saga of Imperial College (UK) COVID model has shown that finally code seen as important. People want to see the code of the model. (That has been used since early 2000s).

Journal initiatives

- Growing appetite among journals (Gigascience, elife, PLOS Comp Bio) with checks of code.
- We have a system CODECHECK for light-touch code review. e.g. Gigascience CODECHECK
- Others more full-on, like CodeOcean.

Other issues that impact on journals.

- Funders are asking for other outputs, beyond traditional paper publication.
- Likewise (in UK) research outputs can include research artifacts. And what the funders ask for, journals I hope will help provide.